

Introduction to Internet of Things(IoT)



Objective:

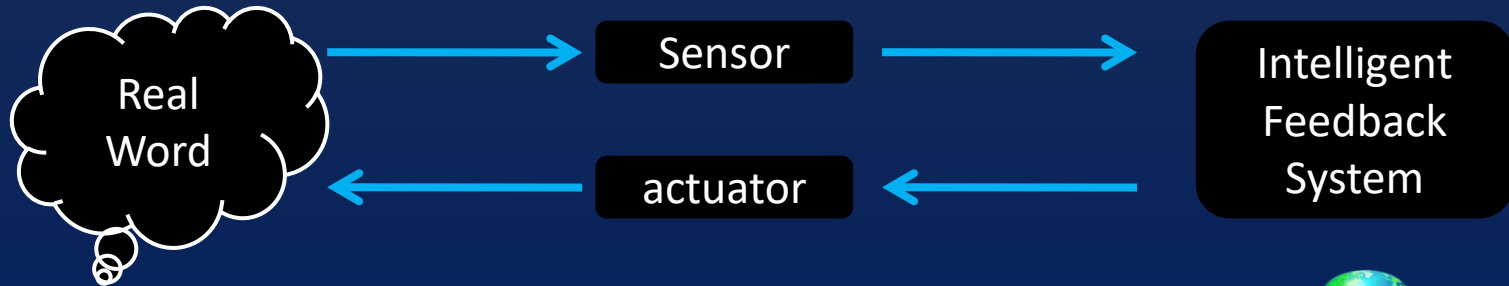
After completing this course, learner will be able to :

- Use different tools for editing image
- Understand different type of image editing
- Working with multiple layers.
- Creating banner, logos, cards and Posters



Introduction

- Sensor and actuators are two critical components of every closed loop system.
- Such a system is also called a mechatronics system.



Sensor

- Sensor are devices that perform input function in a system as they sense the changes in quantity.
- The best example of a sensor is mercury thermometer.
- A sensor is a device which can quantitatively measure a certain physical quantity and converts a physical quantity into an electrical signal.



Classification of Sensor

- All the sensor can be classified into two types based on the power or signal requirement. They are Active sensor and passive sensor.
- Active Sensor:
 - this type of sensor produce output signal with help of external excitation supply. In order to operate active sensor, require power signal from an external source.
 - Strain gauge is an example of active sensor.
- Passive Sensor:
 - it produces output signal without the help of external excitation supply. They do not need any extra stimulus or voltage.
 - A thermocouple is a passive sensor which generates a voltage value corresponding to heat, applied. It does not need any power supply.



Types of Sensor

- **1. Position** : a position sensor measures the position of an object; the position measurement can be either in absolute terms (absolute position sensor) or in relative terms (displacement sensor). Position sensors can be linear, angular, or multi-axis.
- **2. Occupancy and motion** : Occupancy sensor detects the presence of people and animals in surveillance area, while motion sensor detects movement of people and object.
- **3. Velocity and Acceleration** : Velocity (speed of motion) sensor may be linear or angular, indicating how fast an object moves along a straight line or how fast it rotates.



Types of Sensor (con.)

- **4. Force** : Force sensor detects whether a physical force is applied and whether the magnitude of force is beyond a threshold. Example touch sensor.
- **5. Pressure** : Pressure sensor are related to force sensor and measure the force applied by liquids or gases.
- **6. Flow** : Flow sensor detect the rate of fluid flow. They measure the volume (mass flow) or rate (flow velocity) of fluid that has passed through a system in a given period of time.
- **7. Acoustic** : Acoustic sensor measure sound levels and convert that information into digital or analog data signal.



Types of Sensor (con.)

- **8. Humidity** : Humidity is defined as the amount of water present in the surrounding air.
- **9. Light** : Light sensor detects the presence of light. For example infrared sensor, photo detector.
- **10. Radiation** : Radiation sensor detect radiations in the environment. Radiation can be sensed by scintillating or ionization detection.
- **11. Temperature** : Temperature sensor measure the amount of heat or cold that is present in a system.
- **12. Chemical** : Chemical sensor measure the concentration of chemicals in a system.



Types of Sensor (con.)

- **13. Image** : Image sensor are instruments which are used to convert optical image into electronic signal for displaying or storing files electronically. For example thermal image device, radar, sonar, biometric etc.
- **14. Optical** : Optical sensor measure the physical quantity of light rays and covert it into electrical signal which cab be easily readable by user or an electronic instrument/device is called optical sensor. For example fiber optics, Pyrometer, photo detector etc.
- **15. Gas** : gas sensor plays a major role in detecting the gas leakage. These gas sensor are classified into various type based on the type of gas that to be detected.



- **16. Ultrasonic** : ultrasonic means the range of the frequencies. Its range is greater than audible range (>20 kHz) so even it is switched on, you can't sense these sound signals.
- **17. Hall** : Hall sensor is used to detect the magnetic field and gives output in terms of voltage.
- **18. Infrared Sensor (IR)** : it is used to certain characteristics of its surroundings by either emitting or detecting infrared radiation.
- **19. Biosensors** : Biosensor detects various biological elements such as organisms, tissue, cells, enzymes, antibodies,
- **20. Micro and nano-sensors** : micro sensor (sometimes also called MEMS) are the miniaturized version of the conventional macro sensor with improved performance and reduced cost.



Criteria to Choose a Sensor

- Range :
- Resolution
- Accuracy
- Precision
- Sensitivity
- Ruggedness
- Linearity
- Hysteresis
- Response Time



Criteria to Choose a Sensor (Cont.)

- Bandwidth
- Resonance
- Operating Temperature
- Signal-to-noise ratio
- Type of Sensing
- Power Consumption
- Cost
- Repeatability
- Stability



Actuators

- Actuators convert information or energy from a sensor into action by transmitting it to another power mechanism or system, such as heating or cooling a room.
- No human intervention need be involved in the decision-making process.
- Actuators are basically the muscle behind a mechatronics system that accept a control command (mostly in the form of an electrical signal) and produce a change in the physical system by generating force, motion, heat, flow etc.



Classification of Actuators

- Electrical Actuators
- Electromechanical Actuators
- Electromagnetic Actuators
- Hydraulic Actuator
- Pneumatic Actuators
- Smart Material Actuators
- Micro-actuators



Electrical Actuators

- Electrical Actuator is an electromechanical device that converts electrical energy into mechanical that converts electrical energy into mechanical energy .
- Most electric actuators operate through the interaction of magnetic fields and current carrying conductors to generate force.
- Electrical switches are the choice of actuators for most of the on-off type control action.



Electromechanical Actuators

- The most common electromechanical actuator is a motor that converts electrical energy to mechanical motion.
- Motors are the principal means of converting electrical energy into mechanical energy in industry.
- Broadly they can be classified as DC Motors, AC Motors, stepper Motors, and servo motor.



Electromagnetic Actuators

- The solenoid is the most common electromagnetic actuator.
- A DC solenoid actuator consists of a soft iron core enclosed within a current carrying coil.
- When the coil is energized, a magnetic field is established that provides the force to push or pull the iron core.



Hydraulic Actuator

- Hydraulic and pneumatic actuators are normally either rotary motors or liner piston/cylinder or control valves.
- They are ideally suited for generating very large forces coupled with large motion.
- Hydraulic actuators consist of a cylinder or fluid motor that utilize hydraulic power of facilitate mechanical process. Hydraulic motors output liner, rotary, or oscillating motion but acceleration is limited.



Pneumatic Actuators

- Pneumatic actuators work on the same concept as hydraulic actuators except compressed gas is used instead of liquid Energy, in the form of compressed gas, is converted into linear or rotary motion.
- These are used in applications in which positional accuracy is not a requirement.



Smart Material Actuators

- Unlike the conventional actuators, the smart material actuators typically becomes part of the load bearing structures.
- This is achieved by embedding the actuators in a distributed manner and integrating into the load bearing structure that could be used to suppress vibration, cancel the noise, and change shape.



Micro actuators

- Micro actuators, also called Micro-machines Micro-electromechanical system (MEMS). And micro system are the tiny mobile devices being developed utilizing the standard microelectronics processes with the integration of semiconductors and machined micromechanical elements.



Microcontroller

- A Microcontroller is a small computer on a single integrated circuit containing electronic computing unit and logic unit (combinedly known as CPU), Memory (Program Memory and Data Memory), I/O ports and few other components integrated on a single chip.
- Microcontroller is like a mini computer with a CPU along with RAM, ROM, serial ports, timers, and IO peripherals all embedded on a single chip.



Classification of Microcontroller

- Bus width
- Memory Structure
- Instruction set
- Number of input/output Pins



Bus Width

- Bus width indicates the size of the data bus. Higher bus widths often result in better performance.
- Microcontrollers come in 8bit, 16 bit, 32 bit and 64 bit. Some most advanced microcontroller have bits more than 64 which are capable of executing particular functions in the embedded system.
- Based on bit configuration, the microcontroller is further divided into following categories.
 - 8-bit Microcontroller
 - 16-bit Microcontroller
 - 32 –bit Microcontroller



Memory Structure

- Microcontrollers need memory (RAM, ROM, EPROM, EEPROM, Flash memory ,etc) to store programs and data.
- Based on the memory configuration, the microcontroller is further divided into two types :
 - Embedded Memory Microcontroller
 - External Memory Microcontroller



Instruction Set

- Based on the instruction set configuration, the microcontrollers is classification into two categories CISC or RISC.
- CISC : CISC stand for complex instruction set computer. It allows the user to insert a single instruction as alternative to many simple instruction.
- RISC : RISC stands for Reduced Instruction Set Computers. It reduces the operational time by shortening the clock cycle per instruction.



Number of input/output pins

- Microcontrollers vary according to the number of input-output pin sizes.
- One can choose a specific microcontrollers as per the requirement of application.



Component of Microcontroller

- CPU
- Memory
- Input/output ports
- System Bus
- Timers and Counters
- Serial Ports
- Interrupt Control
- ADC (Analog to Digital Converter)
- DAC (Digital to Analog Converter)



Types of Microcontroller

- Microcontroller 8051
- AVR Microcontroller
- PIC Microcontroller
- ARM Microcontroller
- Renesas Microcontroller



Microcontroller 8051

- 8051 microcontrollers is an 8-bit family of microcontroller and is developed by the intel in the year 1981.
- This is one of the popular families of microcontroller being used all across the world.
- It has 128 byte of on chip RAM.
- The data memory can be expanded up to 64kb.
- It is a CISC based Microcontroller with Harvard Architecture (separate program and data memory).



AVR Microcontroller

- AVR Microcontroller was developed in the year of 1996 by Atmel Corporation.
- The structural design of AVR was developed by the Alf-Egil Bogen and Vegard Wollan.
- The AVR microcontroller are modified Harvard RISC architecture with separate memory for data and program.
- The speed of AVR is high when compare to 8051 and PIC.
- The AVR stand for Alf-Egil Bogen and Vegard Wollen's RISC processor.



PIC Microcontrollers

- Peripheral Interface Controller (PIC) is microcontroller developed by a Microchip.
- PIC microcontroller is fast and simple to implement program when you compare other microcontrollers like 8051.
- It Controlled by the software.
- It is RISC based Design
- Its Code os extremely efficient, allowing the PIC to run with typically less program memory than its larger competitors.
- It is low cost, high clock speed.



ARM Microcontroller

- The ARM microcontroller stands for Advance Risk Machine.
- It is one of the extensive and most licensed processor cores in the world.
- The first ARM microcontroller was developed in the year 1978 by Cambridge University.
- The ARM RISC processor was produced by Acorn Group of Computers in the year 1985.
- ARM processors are widely used in customer electronic devices such as smart phones, tablets, multimedia players and other mobile devices such as wearable.



Renesas Microcontroller

- Renesas is latest **automotive** Microcontroller family that offers high performance feature with exceptionally **low power consumption** over a wide and versatile extend of item.
- This microcontroller offers rich functional security and embedded safety characteristics required for new and advanced automotive application..



Embedded System

- An embedded System is a combination of **Hardware and Software**.
- An embedded system can be an independent system or it can be a part of large system.
- It is a microcontroller or microprocessor based on system which is designed to perform **a specific task**.
- It is more of an application oriented system i.e. it is dedicated to perform a **single task**.
- For example Fire alarm is embedded system; it will sense only smoke.



Component of Embedded System

- Processor
- Memory
- Peripherals
- Software



Microprocessor

- A microprocessor is **and Integrated Circuit (IC)** that contains the Central Process Unit (CPU).
- For a microprocessor to work, it needs a bunch of supporting hardware that can be found on a mother board. The hardware includes memory, ICs for peripheral devices, etc.
- The main task of a microprocessor is to accept data as input from input devices then process this data according to the instruction and provide the result of these instruction as output through output devices.
- The first microprocessor was **the Intel 4004**, introduced in **1971**.



Characteristics of Microprocessor

- **Instruction set**: The set of instruction that the microprocessor can execute are called instruction set.
- **Word length** : The number of bits that a computer can manipulate or process at a time is called its word length. The Commonly used word lengths are : 8, 16, 32 or 64 bits.
- **Clock speed** : In a computer, clock speed means the number of pulses per second generated by oscillator that set the number of times the processor processes and operates on data in a second. Clock speed is generally measured in MHz (megahertz or million pulses per second) and GHz (gigahertz or billion pulses per second).



Classification of Microprocessor

- 8-bit Microprocessor
 - Example: Intel 8085
- 16-bit Microprocessor
 - Example: Intel 80286
- 32-bit Microprocessor
 - Example: Intel 80386, 80486, Intel Pentium, Celeron, AMD Sempron
- 64-bit Microprocessor
 - AMD Athlon.



THANK YOU

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